FAHAD AHMED PATWARY

Software Engineer | Java · Python · Full-Stack Web | Software Testing
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PROFESSIONAL SUMMARY

CSE graduate from United International University focused on software engineering and testing. Builds full-stack web apps with Java, Python, React, Next.js, Node.js, and REST APIs, and tests them with Selenium, Postman, and pytest. Strong OOP and data-structures foundation, with research in network security and ML.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, SQL

Core CS: OOP, Data Structures & Algorithms, DBMS, Computer Networks, System Design basics

Web: HTML & CSS, Responsive Design, Bootstrap, Tailwind CSS, React, Next.js, Node.js, REST APIs, GraphQL

Databases: MySQL, MongoDB

Testing: Manual Testing, Test Case Design, Selenium Test Automation, Postman API Testing, pytest, Appium

DevOps & Tools: Linux, Git & GitHub, Docker, Figma (UI design)

Security: Network Security, Cryptography basics, Penetration-testing basics

Machine Learning & Data: PyTorch, LightGBM, XGBoost, ONNX, pandas & NumPy, Excel / Sheets, Power BI / Tableau

Ways of Working: Agile / Scrum, Requirements Analysis, Design Systems, Technical Writing

PROJECT

PustiPath | *Firebase, Next.js, ONNX, Selenium* github.com/FahadPatwary/pustipath

- Designed the architecture of an offline, Bengali-first Android app that screens children for malnutrition with three on-device ONNX models in under 200 ms per screening; built it with AI-assisted coding.
- Exported the trained LightGBM/XGBoost models to ONNX in a 761 KB bundle, validated against the original models with zero output deviation.
- Developed a Firebase backend (Auth, Firestore, Cloud Functions, FCM) with role-based access and default-deny rules, plus a Next.js admin dashboard for users, areas, and health workers.
- Automated unit, RBAC, and end-to-end tests with pytest, Selenium, and Appium, verifying that each user role reaches only its permitted pages.

RESEARCH

Undergraduate Research — United International University 2022– 2026

- **98.5% recall at 0.12 ms per packet:** two-stage intrusion detector (rule engine + One-Class SVM) for IEC 61850 GOOSE smart-grid traffic across five attack types; manuscript,
- **67.9% better Energy-to-Accuracy Ratio, model 255 MB → 91 MB:** Green AI framework combining distillation, pruning, and INT8 quantization on DistilBERT; manuscript
- **ROC-AUC 0.884 on an unseen national holdout:** multimodal child-malnutrition screening with LightGBM, XGBoost, and EfficientNet-B0; manuscript

EDUCATION

B.Sc. in Computer Science & Engineering — United International University (UIU), Dhaka 2022 – [2026]

Relevant Coursework: OOP, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Network Security, Cryptography

Higher Secondary Certificate (HSC) — Birshrestha Munshi Abdur Rouf Public College GPA 5.00 / 5.00 · 2021

Secondary School Certificate (SSC) — Motijheel Model High School GPA 5.00 / 5.00 · 2019